Credit Risk Modeling

Munangiwa Gift Lithole

What is Credit Risk

Credit risk is the risk a bank faces that a borrower may fail to meet their contractual repayment obligations. Since lending is a core activity of banks, this risk is significant because borrower defaults can lead to financial losses, reduced profitability, and lower capital adequacy.

To better understand and manage this risk, banks quantify it using statistical and financial techniques. This process is known as **credit risk modelling**, and it helps banks make decisions such as whether to approve or reject loan applications.

Credit Risk Metrics

There are three key metrics used in credit risk modelling:

- **Probability of Default (PD):** This is the likelihood that a borrower will fail to meet their contractual repayment obligations. A default is commonly defined as the borrower missing payments for a specified period, often 90 days.
- **Exposure at Default (EAD):** This is the total value the bank is exposed to at the time the borrower defaults. It includes the outstanding principal balance, accrued interest, and any applicable fees.
- **Loss Given Default (LGD):** This is the percentage of the Exposure at Default that the bank cannot recover after taking collateral, guarantees, or recoveries into account.

These three metrics can be combined into a single measure known as the **Expected Credit Loss (ECL):**

$$ECL = PD \times EAD \times LGD$$

Conceptually, this formula represents the expected loss a bank may incur from a loan due to borrower default.

Techniques Used in Credit Risk Modeling

Here, we focus on techniques used in modelling **Probability of Default (PD)**.

There are two common types of PD measures: **Lifetime PD** and **12-month PD**. However, our focus is on **12-month PD**, which measures the likelihood that a borrower will default within the next 12 months after the loan has been approved.

The models used for PD estimation are typically **supervised learning** models. This means that we train the models using historical data where the outcomes are already known from past events. Since the objective is to classify borrowers according to risk, PD modelling is generally treated as a **classification problem**, where an application is classified as either a **good borrower** or a **bad borrower**.

Although these models produce probabilities, a threshold is selected to convert the probabilities into classifications. A commonly used default threshold is 0.50. Therefore:

- if the predicted probability of default is greater than or equal to 0.50, the borrower may be classified as a bad borrower;
- otherwise, the borrower is classified as a good borrower.

$$\text{Loan/Application} = \begin{cases} \text{Rejected/Default}, & \text{if } PD \geq 0.5 \\ \text{Approved/No Default}, & \text{if } PD < 0.5 \end{cases}$$

in a way the probalility represent risk, the higher the probability the higher the risk of default

Logistic Regression (Linear Model)

Logistic Regression is a member of the **Generalized Linear Models (GLM)** family, just like Linear Regression.

Both models examine the linear relationship between the **independent variables** (features/inputs) and the **dependent variable** (target/outcome). However, they differ in the type of outcome they predict:

- **Linear Regression** is used when the outcome is a **continuous** variable.
- **Logistic Regression** is used when the outcome is a **categorical** variable (typically binary: 0 or 1).

Logistic Regression predicts the **probability** of an event occurring. This probability, denoted as p, is bounded between **0 and 1**.

If we tried to model p directly using a linear equation like we do in linear regression:

$$p = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

we would run into serious problems. A linear model can produce any value between $(-\infty, +\infty)$, but probability must stay within the range **[0, 1]**. This makes direct modeling of probability inappropriate.

Instead of modeling the probability p directly, we model the **odds**:

$$\text{odds} = \frac{p}{1-p}$$

where p is the probability of the event occurring.

Odds can take any value from **0 to +∞**. However, this range is still not ideal for linear modeling because:

- It is not symmetric,
- It cannot produce negative values.

To solve this, we take the **natural logarithm** of the odds. This transformation gives us the **logit function** (also called log-odds):

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

The logit function has a very useful property: it maps probabilities from (0, 1) to the entire real number line **(-∞, +∞)**. This makes it suitable for linear modeling.

We can now assume a linear relationship between the **log-odds** and the independent variables (features):

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where:

- $x_1, x_2, \dots, x_n$ are the input features,
- $\beta_0, \beta_1, \dots, \beta_n$ are the model coefficients (parameters).

We can rearrange the equation to express the probability \$ p \$ directly:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}}$$

This is known as the **sigmoid function** (or logistic function):

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where $z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$ is the linear combination of the features.

By applying the sigmoid function to the linear predictor, we obtain a valid probability that always lies between **0 and 1**.

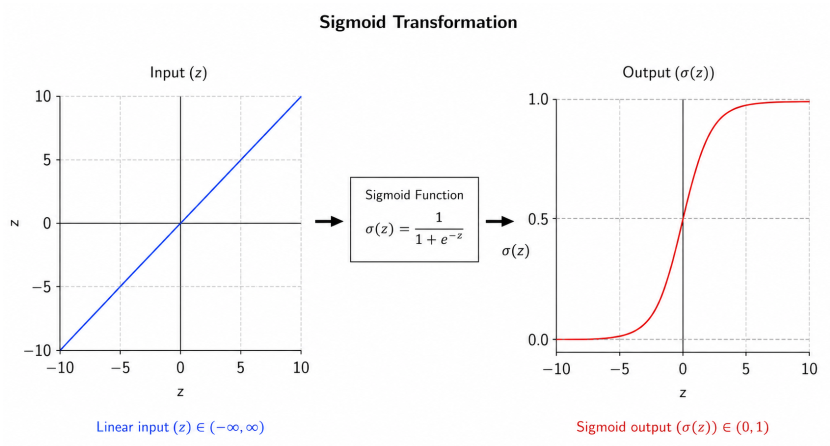

Figure 1: Sigmoid Transformation

Tree-Based Models (Non-Linear)

These are non-parametric supervised learning algorithms that use a hierarchical tree-like structure to make predictions. The algorithms recursively split the features to find optimal split points that best separate the data and improve the prediction of event probabilities.

Tree-based models are particularly useful for capturing non-linear relationships and interactions between variables without requiring strict statistical assumptions.(**No mathematical equations**)

- **Decision Tree**: This model consists of a single tree used to make predictions.
- **Random Forest**: This model consists of many decision trees used for prediction, where the final prediction is typically obtained by averaging the predictions from all the trees.

- **Gradient Boosting Machine**: This model also uses many trees; however, each new tree is built to learn from and correct the mistakes made by the previous trees, improving the overall predictive performance.

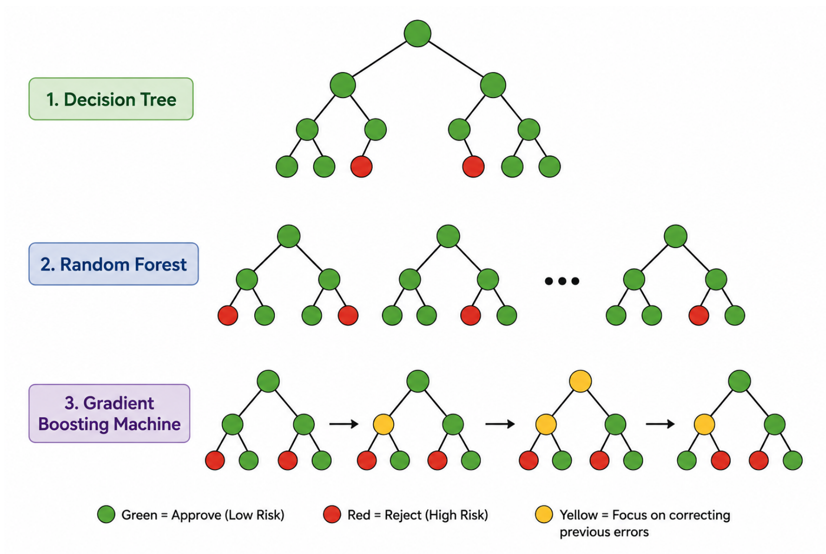

Figure 2: Tree Base Model

Neural Networks (Non-Linear)

A neural network is a machine learning model that stacks simple computational units called *neurons* in multiple layers. These neurons learn patterns in the data by estimating weights and biases that map inputs to outputs.

At a high level, neural networks are inspired by biological neurons in the human brain, which communicate through electrical signals.

The structure of an artificial feed-forward neural network can be represented by the recursive expression:

$$\mathbf{a}^{(l)} = \sigma_l \left(\mathbf{W}_l^T \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad l = 1, 2, \dots, L$$

where:

- $\mathbf{a}^{(l)}$ is the vector of activation values at layer l,
- $\mathbf{W}_l$ is the weight matrix for layer l,
- $\mathbf{b}^{(l)}$ is the bias vector for layer l,
- and σ_l is the activation function applied at layer l.

The initial layer, $\mathbf{a}^{(0)}$, represents the input features for a particular observation, usually denoted by:

$$\mathbf{a}^{(0)} = \mathbf{x}_i$$

Just like in logistic regression, the **activation function** applied at the final layer (L) (the output layer) is typically the sigmoid function when dealing with binary classification problems.

The sigmoid activation function maps the output to a value between 0 and 1, allowing it to be interpreted as a probability of default.

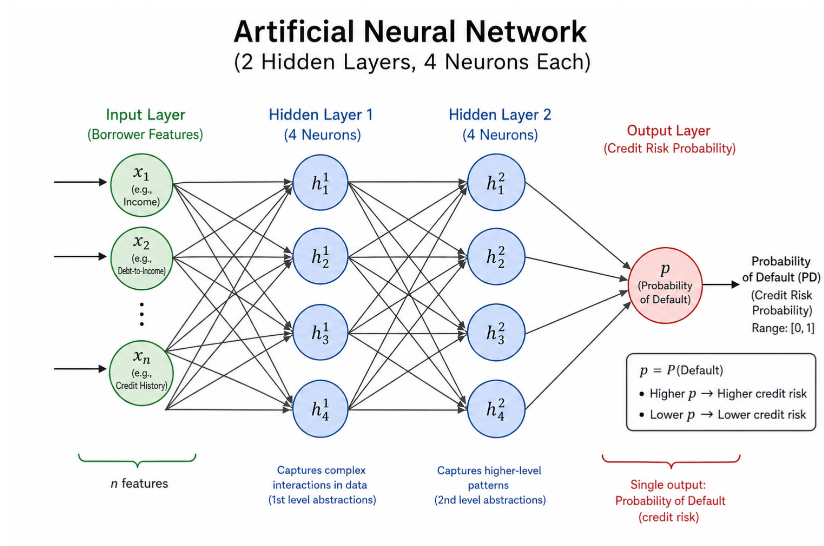

Figure 3: Artificial Neural Network

Others(Non-Linear)

The following are other methods that can also be used:

- Support vector machine
- K-Nearest Neighbors (KNN)
- Naïve Bayes

Complexity and Interpretability

So we can see that logistic regression and decision trees are simple models and do not involve many complicated processes that we can lose track of. This means that these models are interpretable. However, they are not very strong at capturing complex structures.

We also see that in random forests, gradient boosting machines, and neural networks, many processes are involved before reaching a prediction. Therefore, it is easier to lose track of how predictions are made, making these models less interpretable. However, these models are strong at capturing complex structures.

Weights of Evidence(WoE) and Information Value (IV)

- **Weight of Evidence (WoE):** It is a measure of the strength of a feature (independent variable) in separating non-events and events. In our case, it measures how well a feature can separate good and bad loan applications.

$$\text{WoE}_i = \ln\left(\frac{\%\text{ of non-events}_i}{\%\text{ of events}_i}\right) = \ln\left(\frac{\text{Distribution of Good}_i}{\text{Distribution of Bad}_i}\right)$$

i is refer to the ith bin/category

- **Information Value (IV):** It is used to measure the predictive power of independent variables. It helps us understand which independent variables provide the most useful information about the target variable. IV is commonly used as a variable selection technique when the dependent variable (target) is binary, meaning it has only two possible values.

$$\text{IV} = \sum_{i=1}^{n} \left(\text{WoE}_i \times (\%\text{ of non-events}_i - \%\text{ of events}_i)\right)$$

So, **Weight of Evidence (WoE)** is important in credit modelling because we typically use **logistic regression**, which assumes a linear relationship between the **log-odds of the target variable** and the input features.

In practice, WoE is mainly applied to **categorical variables (or continuous variables that have been binned into categories)**. It transforms these categories into numerical values that are linearly related to the log-odds of the target. This makes the features more suitable for logistic regression and improves both model stability and interpretability.

WoE is also useful because it groups categories that have similar distributions with respect to the target variable. This allows us to rank risk across categories by observing how the default rate changes from one group to another.

Ideally, we want this relationship to be **monotonic** (either consistently increasing or consistently decreasing), meaning that risk should move in a single direction across the ordered categories. This makes the model more interpretable and aligned with credit risk intuition.

Model Evaluation Metrics

So, the **positive class** is the event we are modelling, which is a bad loan application (default), and the **negative class** represents the non-event, which is a good loan application (non-default).

Many metrics that follow are defined from the **confusion matrix**:

- **True Positive (TP):** The actual event is default, and the model predicts default.
- **False Negative (FN):** The actual event is default, however the model predicts non-default.
- **False Positive (FP):** The actual event is non-default, however the model predicts default.
- **True Negative (TN):** The actual event is non-default, and the model predicts non-default.

	Predicted Positive	**Predicted Negative**
Actual Positive	TP	FN
Actual Negative	FP	TN

Table 1: Confusion Matrix

1. **Accuracy**: This is the overall percentage of correct predictions. It measures how many of the total predictions made by the model were correct, whether default or non-default.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

2. **Precision**: This measures how well the model predicts defaults (the positive class). Out of all loan applications predicted to default, it shows how many actually defaulted.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

3. **Recall**: This measures how well the model identifies all actual default events. Out of all loan applications that actually defaulted, it shows how many the model correctly predicted as default.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

4. **F1 Score**: This is the harmonic mean of Precision and Recall. It is used to balance the trade-off between the two. For example, if one model has high precision and another has high recall, the F1 score helps identify a balance between them and can be used to compare models.

5. **AUC (Area Under the Curve)**: This refers to the area under the Receiver Operating Characteristic (ROC) curve. The ROC curve shows how well the model separates default (events) from non-default (non-events) across all possible classification thresholds used to define an event.

6. **Gini**: This also measures how well the model separates defaults and non-defaults. It is directly related to AUC:

$$\text{Gini} = 2 \times \text{AUC} - 1$$

Forbidden Features

The features that a regulator will disapprove of are primarily those that introduce discrimination and those that lack business interpretability.

Forbidden features include race, religion, gender, marital status, etc. They also include **proxy variables** of such attributes—features that are indirectly correlated with protected characteristics, such as zip code, geographical data, email domain, etc.

These features are disallowed because of fair lending practices. Models must not perpetuate bias or discrimination, and they must also support explainability and accountability, since lenders are required to provide clear reasons when rejecting a loan application.

Data Cleaning Approach

The dataset (loan book) contains **602 duplicate records**, which will be removed.

There are **4 variables** with missing values:

- annual_income: 8638
- employment_length_years: 3711
- num_open_accounts: 2423
- months_since_last_delinquency: 60068

The variable **months_since_last_delinquency** was dropped because nearly half of its observations contain missing values.

For **annual_income**, the data was grouped by region, and the missing values were imputed using the median annual income within each region. The reason for this approach is that income levels may differ across locations.

For **employment_length_years** and **num_open_accounts**, missing values were imputed using the median of each variable.

The variables **age**, **employment_length_years**, **num_open_accounts**, and **months_since_oldest_account** had float data types, although they should be stored as integers. In addition, **application_date** had a string (object) data type instead of a date format.

The data types were converted to their appropriate formats.

It was also observed that the variables **home_ownership** and **loan_purpose** contained inconsistent label formatting. The category labels were standardized to ensure consistency.

Exploratory Data Analysis(EDA)

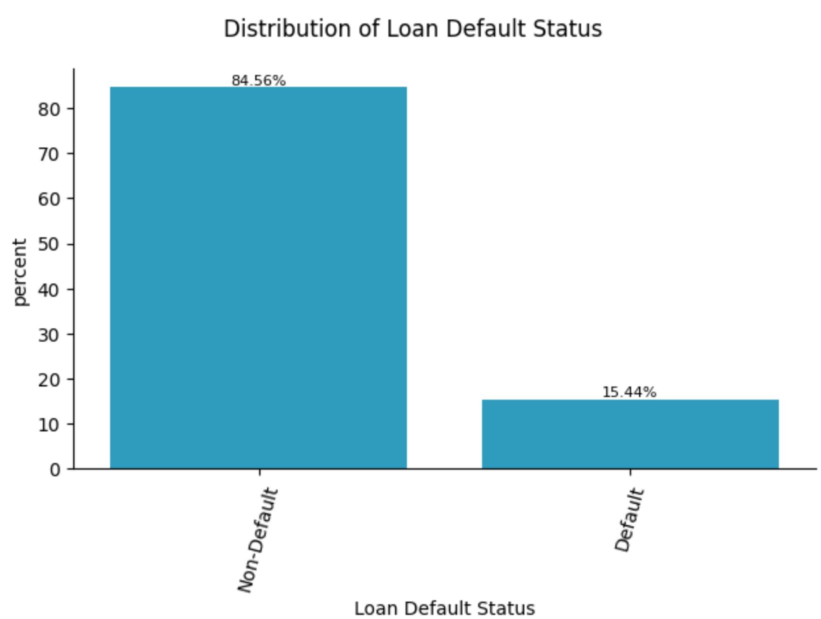

Figure 4: Target Distribution

We start by looking at the **target distribution**. From the target distribution diagram, we can see that people who defaulted make up 15.44% of the dataset, while non-defaults account for 84.56%. This shows that the dataset is imbalanced and further implies that there are many good customers and relatively few bad customers.

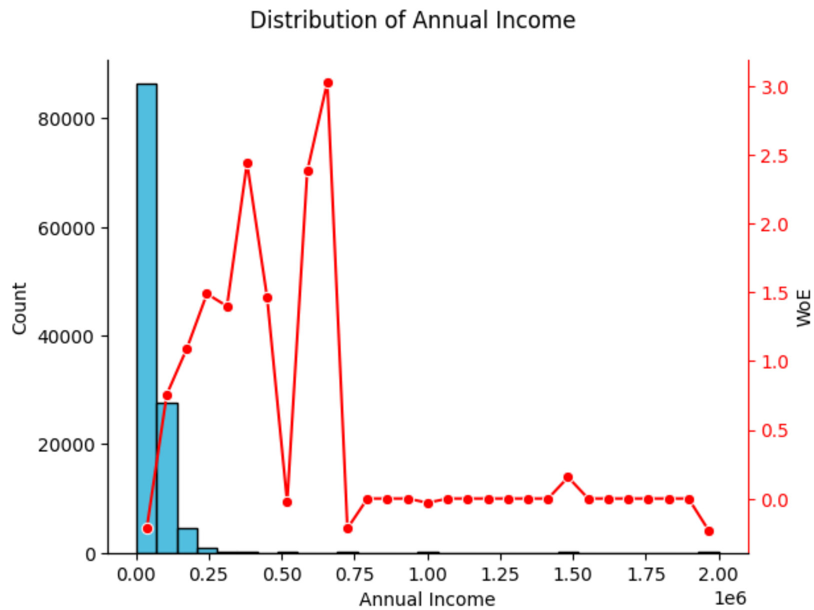

Figure 5: Annual Income

Next, we consider the histogram of annual income. From the histogram, we can observe that there are many tall bars on the left side and fewer bars spread across the right side. This indicates that annual income is right-skewed (has a long right tail) and may contain potential outliers. This suggests that most people earn lower incomes, while only a few people earn very high incomes.

Now consider the WoE plot (shown in red). We can observe that the red line generally increases from one bin to another, indicating that the risk of default decreases as annual income increases. Although the relationship is not perfectly monotonic, this is likely because some bins at the higher income levels contain very few customers. For example, if a bin contains only one customer and that customer defaulted, the WoE value may fluctuate significantly. This explains why the red line becomes unstable in the later bins.

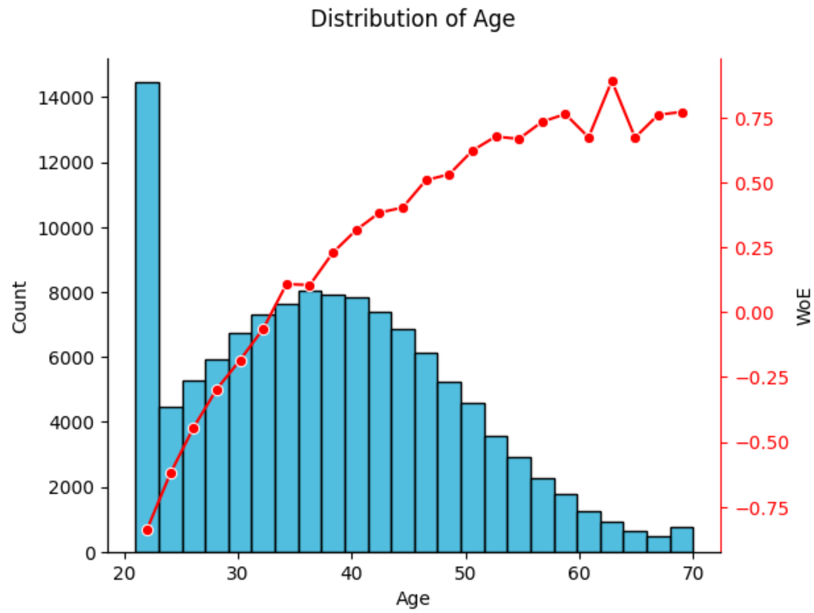

Figure 6: Age

The histogram for age appears approximately normally distributed, with a larger concentration of younger applicants around age 21. The WoE plot (red line) is consistently increasing, showing the monotonic relationship that we expect. However, the curve increases at a decreasing rate, suggesting a non-linear relationship between age and the log-odds of default.

The increasing WoE trend indicates that as age increases, the risk of default decreases. This suggests that younger applicants are generally more likely to default than older applicants.

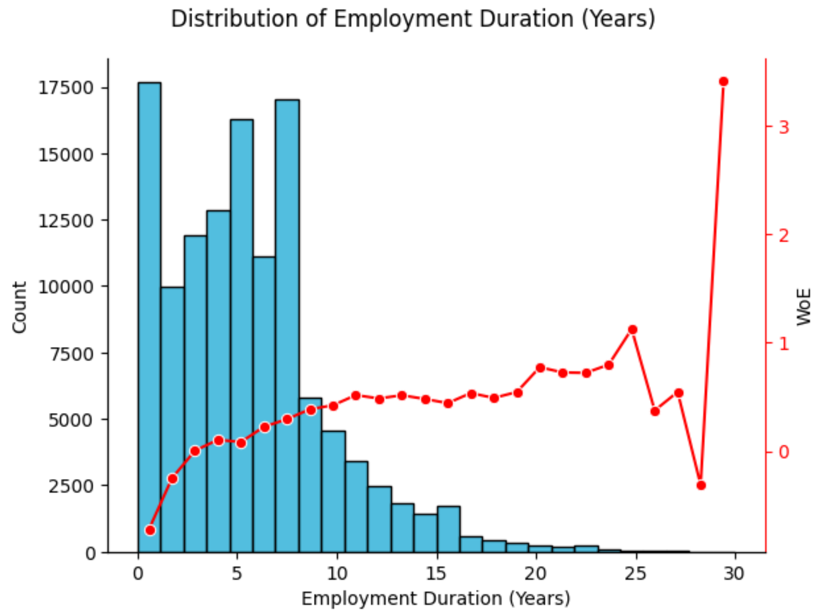

Figure 7: Employment Duration (Years)

For employment duration, the histogram shows a slight right tail, with most people having employment durations between 0 and 10 years. The WoE plot (red line) generally increases, although it is not perfectly monotonic. The curved shape suggests that employment duration is not perfectly linearly related to the log-odds.

The increasing trend implies that as employment duration increases, the risk of default decreases. This suggests that early-career professionals are generally more likely to default compared to more experienced individuals.

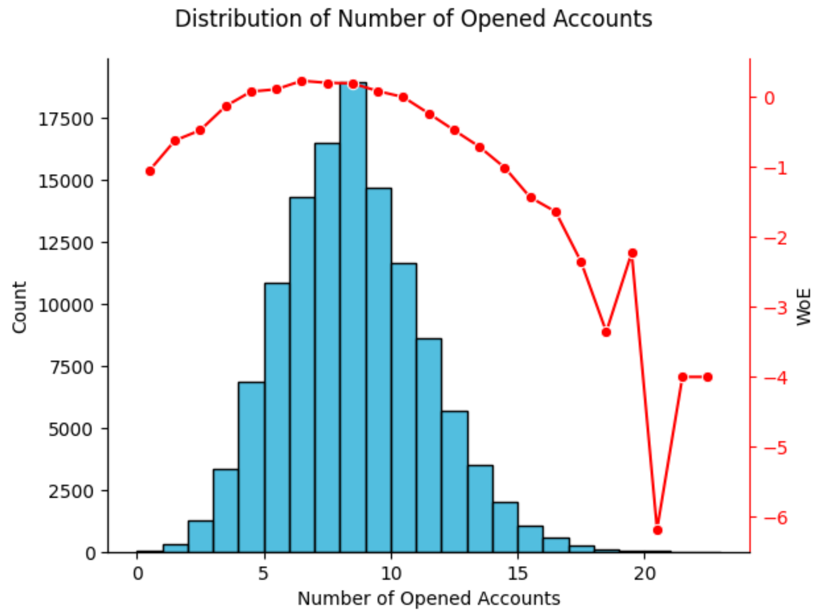

Figure 8: Number of Opened Accounts

The histogram for the number of opened accounts appears approximately normally distributed, with most people having around 8 opened accounts. The WoE plot forms a quadratic-like shape, suggesting that the variable does not have a linear relationship with the log-odds, but rather a non-linear relationship.

This indicates that when customers have very few accounts, opening additional accounts may initially reduce risk. However, after reaching approximately 8 accounts, the risk of default begins to increase again as more accounts are opened. This suggests that both very low and very high numbers of opened accounts may be associated with higher default risk, although having too many accounts appears riskier.

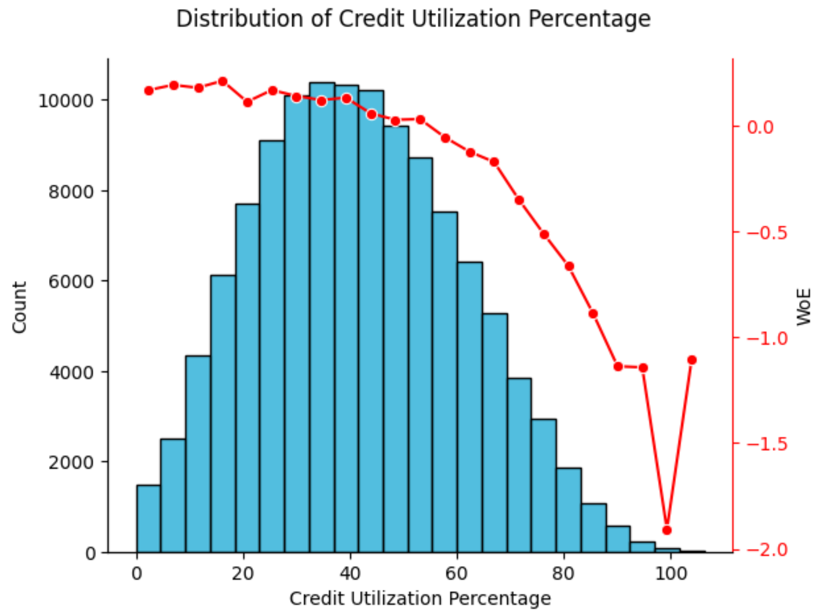

Figure 9: Credit Utilisation Percentage

The distribution of credit utilisation percentage appears approximately normally distributed, with many customers using around 41% of their available credit. The WoE plot is consistently decreasing, showing a monotonic relationship, although it is not perfectly linear because the curve is slightly curved.

The decreasing WoE trend indicates that as credit utilisation increases, the risk of default also increases. This implies that customers who use a smaller proportion of their available credit are generally less risky than those with high credit utilisation.

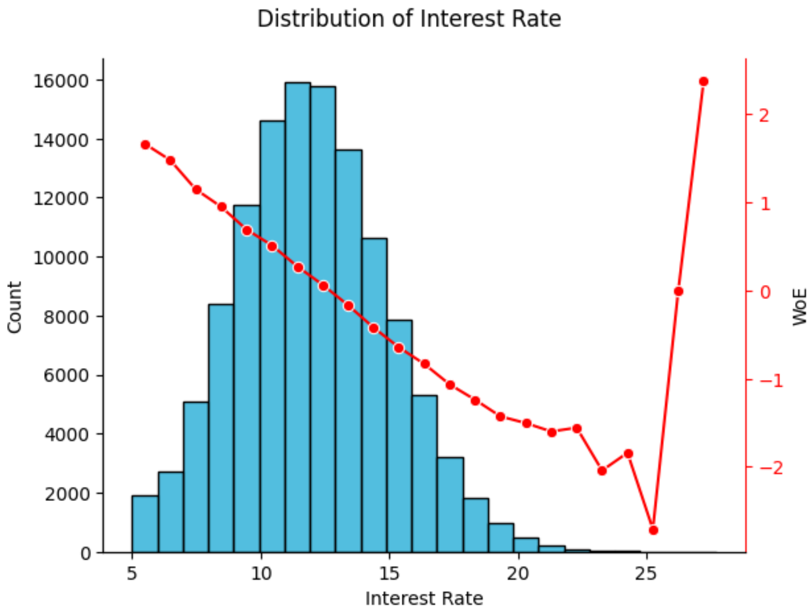

Figure 10: Interest Rate

For interest rate, the distribution appears approximately normal, with most customers being charged around a 12% interest rate. The WoE plot generally decreases, which is the monotonic relationship we expect, although there is some zig-zag behaviour at the extreme bins that may be due to potential outliers.

This suggests that interest rate has a relatively linear relationship with the log-odds of default. We can further conclude that higher interest rates are associated with a higher risk of default. In other words, when loans become less affordable, borrowers are more likely to default.

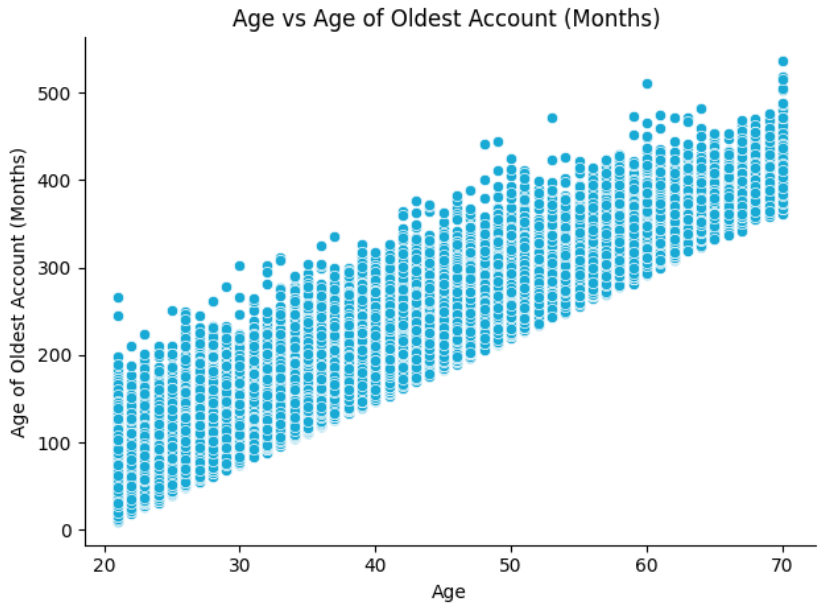

Figure 11: Age vs Months Since The Oldest Account

Age is positively and linearly related to months since the oldest account. This means that older individuals tend to have older credit accounts. The relationship suggests a strong correlation between the two variables.

Feature Engineering Choices

For feature engineering, we first consider **annual_income**. Since the variable was right-skewed and contained potential outliers, a new variable called `log_annual_income_capped` was created. First, the `annual_income` variable was capped to reduce the effect of extreme outliers, and then a logarithmic transformation was applied to the capped values. The logarithmic transformation helps improve the relationship between annual income and the log-odds by making the relationship more linear.

For the variables `age`, `home_ownership`, `num_open_accounts`, `credit_utilisation_pct`, and `employment_length_years`, the exploratory data analysis showed that these features had non-linear relationships with the log-odds of default. To better capture these relationships, new transformed variables ending with `_WoE` were created, for example `feature_WoE`.

For each variable, the values were separated into bins, and each observation was replaced with the corresponding Weight of Evidence (WoE) value for the bin to which it belonged. This transformation helps capture non-linear relationships while also making the variables more suitable for logistic regression.

Modeling Methodology

For this credit risk model, logistic regression was selected as the primary modelling technique. The main reason for choosing logistic regression is that it is highly interpretable and suitable for binary classification problems, where the objective is to distinguish between good and bad customers by predicting the likelihood of default.

In addition, logistic regression produces predicted probabilities, which can be used to support lending decisions such as whether to approve or reject a loan application.

The dataset was split into training and testing sets using the variable named `set` in the data. The training set contains 70% of the observations, while the testing set contains the remaining 30%.

A baseline logistic regression model was first fitted after removing all forbidden features. Thereafter, additional engineered features were added one at a time to evaluate whether they improved model performance. This process continued until all newly engineered features had been assessed.

The primary objective during model development was to improve the model's **AUC (Area Under the ROC Curve)**, since AUC measures how well the model distinguishes between default and non-default borrowers.

Final Logistic Regression Model

The final logistic regression model obtained is and variables where scaled:

$$
\begin{aligned}
\log\left(\frac{p}{1-p}\right) = & -0.4318 \\
& -0.5754(\text{log\_annual\_income\_capped}) \\
& +0.5387(\text{num\_delinquencies\_2yr}) \\
& -0.3895(\text{age\_woe}) \\
& -0.3275(\text{num\_open\_accounts\_woe}) \\
& +0.2715(\text{num\_hard\_inquiries\_6mo}) \\
& -0.2678(\text{credit\_utilisation\_pct\_woe}) \\
& -0.2371(\text{employment\_length\_years\_woe}) \\
& +0.1012(\text{dti\_ratio}) \\
& -0.0907(\text{pct\_accounts\_current}) \\
& +0.0835(\text{interest\_rate}) \\
& -0.0430(\text{home\_WoE}) \\
& +0.0262(\text{loan\_amount})
\end{aligned}
$$

The estimated intercept of the model is:

$$\beta_0 = -0.4068$$

The odds ratios corresponding to each coefficient are shown below:

Feature	**Coefficient**	**Odds Ratio**
log_annual_income_capped	−0.5754	0.562
num_delinquencies_2yr	0.5387	1.714
age_woe	−0.3895	0.677
num_open_accounts_woe	−0.3275	0.721
num_hard_inquiries_6mo	0.2715	1.312
credit_utilisation_pct_woe	−0.2678	0.765
employment_length_years_woe	−0.2371	0.789
dti_ratio	0.1012	1.107
pct_accounts_current	−0.0907	0.913
interest_rate	0.0835	1.087
home_WoE	−0.0430	0.958
loan_amount	0.0262	1.027

Interpretation of the Final Logistic Regression Model

Since the variables were scaled before fitting the logistic regression model, the coefficients represent the effect of a one standard deviation increase in each feature on the log-odds of default, holding all other variables constant.

The odds ratios describe the multiplicative change in the odds of default associated with a one standard deviation increase in a feature.

- **log_annual_income_capped** has a negative coefficient (-0.5754) and an odds ratio of 0.562. This indicates that higher income is associated with lower default risk. A one standard deviation increase in this variable decreases the odds of default by approximately 43.8%.
- **num_delinquencies_2yr** has a positive coefficient (0.5387) and an odds ratio of 1.714. This suggests that borrowers with more past delinquencies are more likely to default. A one standard deviation increase in this feature increases the odds of default by approximately 71.4%.
- **age_woe** has a negative coefficient (-0.3895), indicating that borrowers in lower-risk age groups are generally less likely to default.
- **num_open_accounts_woe** has a negative coefficient (-0.3275), suggesting that favourable account structures are associated with lower default risk.
- **num_hard_inquiries_6mo** has a positive coefficient (0.2715) and an odds ratio of 1.312. This indicates that borrowers with more recent hard inquiries are more likely to default.
- **credit_utilisation_pct_woe** has a negative coefficient (-0.2678), showing that lower-risk credit utilisation groups are associated with lower default probabilities.
- **employment_length_years_woe** has a negative coefficient (-0.2371), suggesting that stronger employment profiles are associated with lower default risk.
- **dti_ratio** has a positive coefficient (0.1012), indicating that higher debt-to-income ratios are associated with increased default risk.
- **pct_accounts_current** has a negative coefficient (-0.0907), implying that borrowers with a higher proportion of current accounts are generally less risky.
- **interest_rate** has a positive coefficient (0.0835), suggesting that higher interest rates are associated with higher default risk.
- **home_WoE** has a small negative coefficient (-0.0430), indicating a relatively weak relationship with default risk.
- **loan_amount** has a positive coefficient (0.0262), suggesting that larger loan amounts slightly increase default risk.

Performance Relative to Baseline

The current logistic regression model achieved an AUC of 0.72, outperforming the baseline logistic regression model (AUC = 0.68).

- **Absolute improvement**: +0.04
- **Relative improvement**: +5.9% over the baseline